Vishhvaan Gopalakrishnan

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EDUCATION

CLEVELAND CLINIC
CASE WESTERN RESERVE U.
MEDICAL DOCTOR (M.D.)
W/ RESEARCH QUALIFICATIONS
May 2020 | Cleveland, OH

UNIVERSITY OF MICHIGAN AT ANN ARBOR
M.S.E. AND B.S.E. IN BIOMEDICAL ENGINEERING
May 2013 | Ann Arbor, MI
Summa Cum Laude

MEDICAL TRAINING

ROTATIONS

AI: Internal Medicine • Pediatrics
Core: Surgery • Family Medicine • Neurology • OB/GYN • Psychiatry • Emergency Medicine
Electives: Dermatology • Radiology • Palliative Medicine/Hospice • Cardiology Consults • Cardiac Imaging • Infectious Diseases
(AI is Acting Intern meaning full team responsibility)

COMPUTING SKILLS

PROGRAMMING

Over 5000 lines:
Shell • Python • R • MATLAB
HTML • CSS • LaTeX
Familiar: Go • C++
Learning: JavaScript • SQL

APPLICATIONS

Docker/Kubernetes • TensorFlow
KiCAD • Git • PGP • NGINX • Hugo
Vim • VMWare ESXi

LINKS

Website	**vishhvaan.com**
Github	**vishhvaan**
Twitter	**@vishhvaan**
LinkedIn	**vishhvaan**
Keybase	**vishhvaan**

RESEARCH

THEORY DIVISION (EVOLUTION AND DRUG RESISTANCE) | POST-DOC FELLOW
Dec 2018 - Present | Cleveland, OH
Lead the creation of an open source bioreactor to studies evolution of cultured bacteria and cancer. This tool informs clinical decision making about patient-specific optimal drug therapies. Work in the Department Translational Hematology & Oncology Research with **Jacob Scott, M.D.**.

HEART AND VASCULAR INSTITUTE GENOMICS | RESEARCHER
Jul 2015 – Jul 2017 | Cleveland, OH
Investigated genomic etiologies for the incidence differences in atrial fibrillation between men and women with **Mina Chung M.D.** in Department of Cardiac Electrophysiology. Performed analysis in R and basic machine learning in Python (scikit-learn).

THERAPEUTIC ULTRASOUND GROUP | MASTER'S THESIS PROJECT
Jan 2012 – Apr 2013 | Ann Arbor, MI
Designed and prototyped an acoustic coupler for a clinical procedure (histotripsy) with my team in the **Xu lab**. We programmed (MATLAB) control of the 3D movement and orientation of the coupler. Work led to designs that were used for patent applications by the lab.

AWARDS

2019	1/15 Nationally	Hematology Opportunities for the Next Generation of Research Scientists (**HONORS**) by The American Society of Hematology
2013	32 Medical Students	Cleveland Clinic Medical School **Full Tuition Scholarship**
2012	University of Michigan	University of Michigan **Honors** College of Engineering **Dean's List**

EXPERIENCES

LIVESCRIBE INC. | RESEARCH AND DEVELOPMENT ENGINEERING INTERN
June 2010 – Sep 2010 | Oakland, CA

- Developed cloud-based sync software for the Livescribe smartpen and performed market research of back-end cloud hosting solutions.
- Learned about database management and server-client relationships.

SIEMENS INC. | RESEARCH AND DEVELOPMENT ENGINEERING INTERN
June 2009 – Sep 2009 | Buffalo Grove, IL

- Developed the circuits of a controller for new rainwater catchment product. This product help facilities improve their eco-footprint (LEED guidelines).
- Gained experience with circuit and printed circuit board (PCB) design along with bus communication.

ACTIVITIES

- CWRU Free Clinic ..... Business Leader
- Peru Health Outreach Program – CWRU Global Health Program ..... Overall Leader
- Med. School Tech Committee ..... Member
- M-HEAL (Eng. Outreach) .... Member
- U. Mich. Engineering Council ..... Leadership
- U. Mich. Math Lab ..... Teacher

TECH IN MEDICINE

Areas I can develop both as a programmer and as a doctor:

- Improving inpatient and surgical workflow/handoff.
- Improving the Electronic Medical Record's (EMR) user interface and helpfulness.
- Designing useful biometric markers for continuous health monitoring and analysis.
- Designing and integrating computer vision into medical imaging and machine learning models into the EMR.

REFERENCE

Jacob Scott, M.D., DPhil

Superior for almost 2 years

216·903·1517
scottj10@ccf.org

"He is a self-starter, and is often found in lab late at night and on the weekends, doggedly pursuing this project. It is a rare student that can manage to master so many disciplines."

PROJECTS

OPEN SOURCE EVOLUTIONARY BIOREACTOR | Engineering

github.com/vishhvaan/eve-pi

- Defines an open framework for an automated continuous culture system.
- This project incorporates hardware, I^2C-based circuitry, and PCB designs, a Python-based software package, and experimental protocols, and sample experimental data.
- Software is forked and used by researchers at the Children's Hospital of Philadelphia, University of Cape Town, BforCure (a French BioTech company), and others.

IMMEDIATE RECURRENCE OF ATRIAL FIBRILLATION SUSCEPTIBILITY | Machine Learning

github.com/vishhvaan/ml-iraf-analysis

- This project aims to form better predictions about when patients are at risk for recurrence of atrial fibrillation after cardioversion.
- I use a Cleveland Clinic cardioversion database in combination with XGBoost and Deep Learning techniques.
- Models are queried for insights that could shed light on the pathophysiology of recurrence which could be tested in follow-up biological or clinical research.

DIY INTERNET OF THINGS THERMOSTAT | Engineering

github.com/vishhvaan/thermostat

- Enables users to make use of cheap temperature (and humidity, barometric pressure, and volatile organic compound gas) sensors.
- An application written in Go and executed on the Raspberry Pi processes sensor-generated data.
- Data is read by a Home Assistant instance to automate functions around the user's home.

PUBLICATIONS

[1] V. Gopalakrishnan, K. J. Card, D. Crozier, A. Dhawan, M. Dinh, E. Dolson, N. Farrokhian, E. Ho, T. Jagdish, E. S. King, N. Krishnan, G. Kuzmin, J. Maltas, J. Mo, J. Pelesko, J. A. Scarborough, J. G. Scott, G. Sedor, E. Tian, and D. T. Weaver. UV Sterilization of Personal Protective Equipment with Idle Laboratory Biosafety Cabinets During the COVID-19 Pandemic. *bioRxiv*, 2020.

[2] V. Gopalakrishnan, N. P. Krishnan, E. McClure, J. Pelesko, D. Crozier, D. F. Williamson, N. Webster, D. Ecker, D. Nichol, S. Mandal, R. A. Bonomo, and J. G. Scott. A Low-Cost, Open Source, Self-Contained Bacterial EVolutionary biorEactor (EVE). *bioRxiv*, 2019.

[3] B. J. Kayani, D. T. Weaver, V. Gopalakrishnan, E. S. King, E. Dolson, N. Krishnan, J. Pelesko, M. J. Scott, M. Hitomi, J. G. Scott, and I. Charnas. Assessment of N95 respirator decontamination and re-use for SARS-CoV-2. *medRxiv*, 2020.